Separable 1st order ODE

$$\frac{dy}{dx} = g(x) \cdot p(y)$$

Ex: $\frac{dy}{dx} = \frac{2x+xy}{1+y^2} = x \cdot \frac{2+y}{1+y^2}$ separable

$\frac{dy}{dx} = 1+xy$ not separable

Method to solve: $\frac{dy}{p(y)} = g(x)dx$

$\overset{\text{integrate}}{\Longrightarrow} \int \frac{dy}{p(y)} = \int g(x)dx \Rightarrow H(y) = G(x) + C$

Note: If $p(r) = 0$ then $y \equiv r$ is also a solution.

Ex: $\frac{dy}{dx} = 3y^{2/3}$ - separable $\underset{y \neq 0}{\Rightarrow}$ $\frac{dy}{y^{2/3}} = 3dx \Rightarrow$

Consider:

$\boxed{y \equiv 0}$

solution

$\Rightarrow \int \frac{dy}{y^{2/3}} = \int 3dx \Rightarrow$

$\Rightarrow \boxed{3y^{1/3} = 3x + C_1}$ - general implicit solution

$\Rightarrow y^{1/3} = x + C$

$\Rightarrow \boxed{y = (x+c)^3}$ - general explicit solution